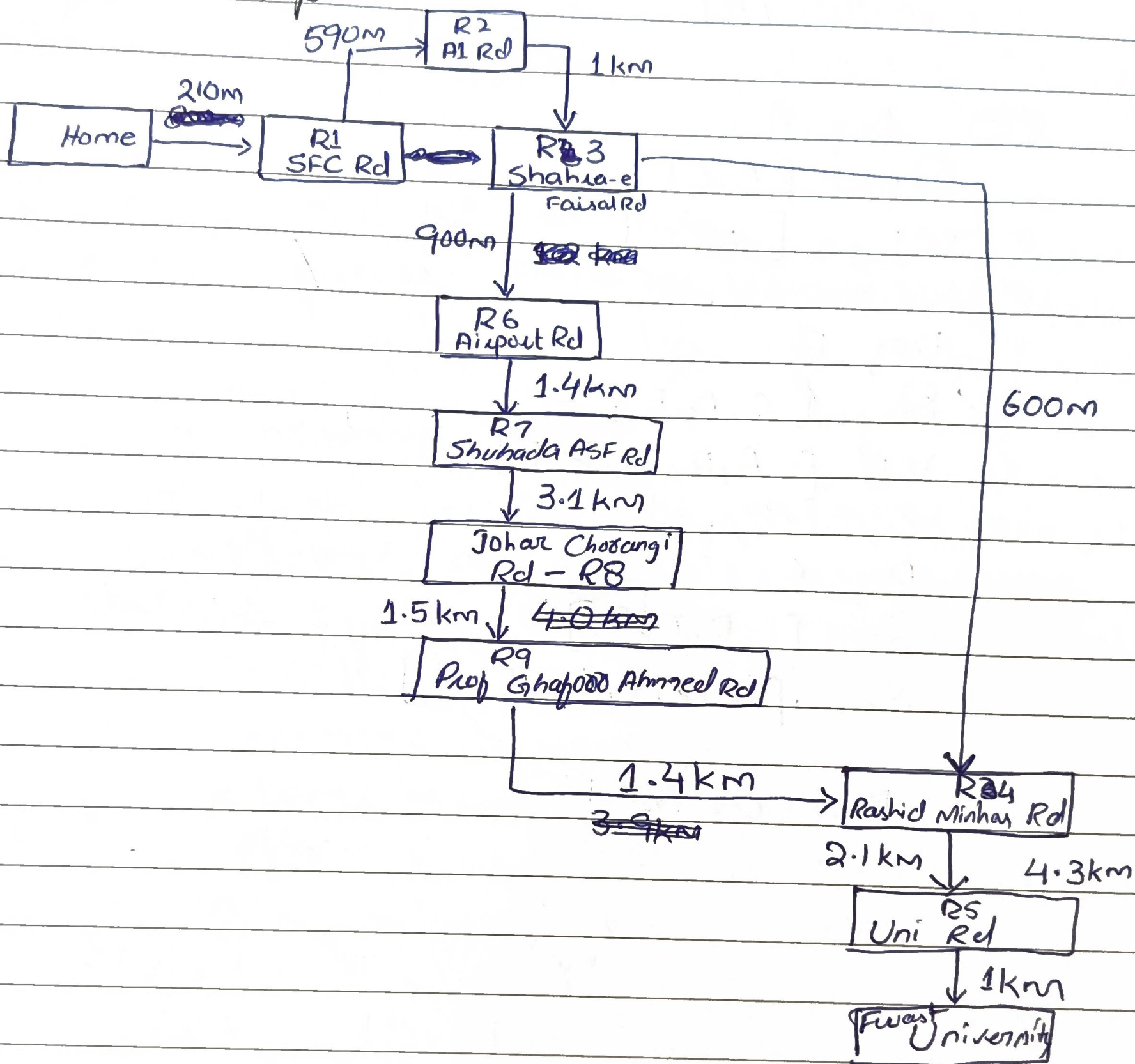


Home To Uni Route BFS

Date:

1. Route Map



Route 1: ~~210 + 590~~

$$0.21 + 0.59 + 1 + \cancel{0.9} + 0.6 + 4.3 + 1 = 7.7 \text{ km}$$

Route 2:

$$0.21 + 0.59 + 1 + 0.9 + 1.4 + 3.1 + 1.5 + 1.4 + 2.1 + 1 = 13.2 \text{ km}$$

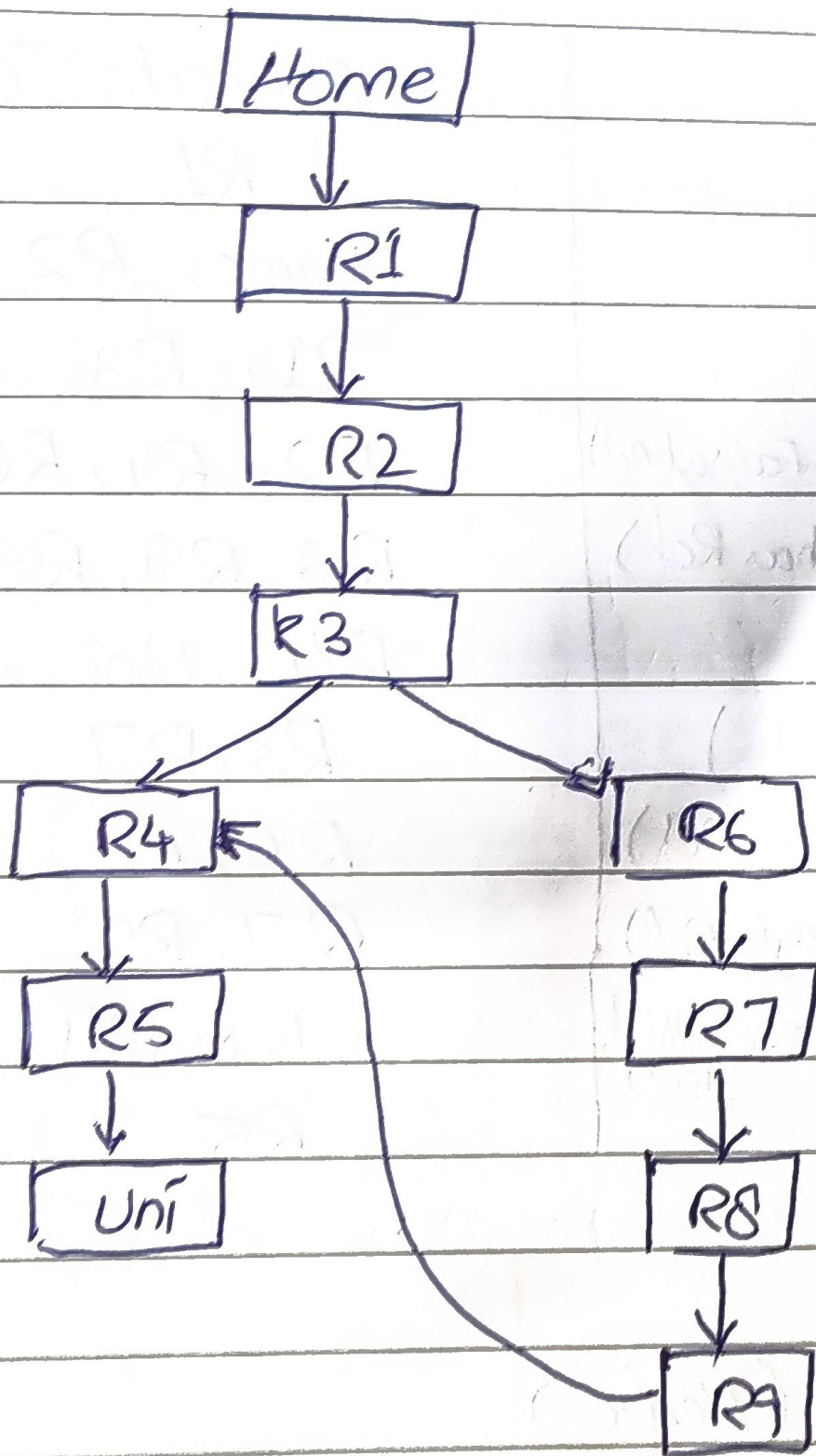
Brilliant

2. Graph Representation:

Node	Connects To
Home	R1
R1 (SFC Rd)	Home, R2
R2 (A-1 Rd)	R1, R3
R3 (Shahrah-e-Faisal Rd)	R2, R4, R6
R4 (Rashid Minhas Rd)	R3, R9, R5
R5 (Uni Rd)	R4, Uni
R6 (Airport Rd)	R3, R7
R7 (Shuhada ASF Rd)	R6, R8
R8 (Johar chowangi Rd)	R7, R9
R9 (Prof. Ghaffar Ahmed Rd)	R8, R4
FUWAST Uni	R5

3. BFS Tree

3. BFS TREE:



4. BFS Trace:

Step 0: Initialization

Q = { Home }

V = { Home }

S1: deg ~~R1~~ Home
 $Q = \{R1\}$
 $V = \{Home, R1\}$

Home $\xrightarrow{Base}$ (R1)
~~expand R1 $\rightarrow$ Home~~
~~Expand Home~~

S2: deg R1
 $Q = \{R2\}$
 $V = \{Home, R1, R2\}$

Expand R1 $\rightarrow$ (Home, R2)
 Home already visited

S3: deg R2
 $Q = \{R3\}$
 $V = \{Home, R1, R2, R3\}$

Expand R2 $\rightarrow$ (R1, R3)
 R1 already visited

S4: deg R3
 $Q = \{R4, R6\}$
 $V = \{Home, R1, R2, R3, R4, R6\}$

Expand R3 $\rightarrow$ (R2, R4, R6)
 R2 already visited

S5: deg R4
 $Q = \{R6, R9, R5\}$
 $V = \{Home, \dots, R4, R9, R5\}$

Expand R4 $\rightarrow$ (R3, R9, R5)
 R3 already visited

S6: deg R6
 $Q = \{R9, R5, R7\}$
 $V = \{ \dots, R6, R7 \}$

Expand R6 $\rightarrow$ (R3, R7)
 R4 already visited

7: Deq R9 $\rightarrow$ Expand R9 $\rightarrow$ (R8, R4)
R4 already visited

$$Q = \{R5, R7, R8\}$$

$$V = \{ \text{---}, R9, R8 \}$$

8: Deq R5 $\rightarrow$ Expand R5 $\rightarrow$ (R4, Uni)
R4 already visited

$$Q = \{R7, R8, \text{Uni}\}$$

$$V = \{ \text{---}, R5, \text{Uni} \}$$

9: Deq R7 $\rightarrow$ Expand R7 $\rightarrow$ (R6, R8)

R6 already visited, R8 already visited

$$Q = \{R8, \text{Uni}\}$$

10: Deq R8 $\rightarrow$ Expand R8 $\rightarrow$ (R7, R9)

Both are visited

$$Q = \{ \text{Uni} \}$$

11: Deq Uni $\rightarrow$ Goal Found.

BFS order of expansion.

Home $\rightarrow$ R1 $\rightarrow$ R2 $\rightarrow$ R3 $\rightarrow$ R4 $\rightarrow$ R6 $\rightarrow$ R9,
 $\rightarrow$ R5 $\rightarrow$ R7 $\rightarrow$ R8 $\rightarrow$ Uni

Shortest Route:

Home $\rightarrow$ R1 $\rightarrow$ R2 $\rightarrow$ R3 $\rightarrow$ R4 $\rightarrow$ R5
 $\rightarrow$ Uni
(6 steps)